

PULCHOWK ENGINEERING CAMPUS TRIBHUVAN UNIVERSITY



Project Report on “CV DATABASE”

Submitted To:

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ABSTRACT

This report describes the implementation of a Database Management System for CV Management (CV-MDS). It was developed for storing, managing, and updating the professional data that job applicants and recruitment departments generate. This report mainly includes the background and development process of the project. The main steps in the development process are identifying the entities (such as Users, Education, Experience, and Skills) and the relationships among them, and presenting them in graphical form with an ER diagram. It also includes the conversion of the conceptual model into a relational model, the application of normalization (up to 3NF) to ensure data integrity, and some of the SQL queries used in the implementation of the MySQL Database. This project is also useful for HR departments, recruitment agencies, and other similar organizations looking to transition from traditional document-based storage to a structured relational database.

ACKNOWLEDGEMENT

We are very pleased to present the report on “CV Management Database System” which has been prepared for the partial fulfillment of the requirements for the degree of Bachelor in computer Engineering(BCT).

We would like to express our gratitude and sincere thanks to our teacher [Insert Teacher's Name] for his/her guidance and support in the preparation of our report. We would also like to extend our gratitude to all of them who have helped us during the preparation of our report, i.e., our friends and family. Finally, we would like to acknowledge our college management for providing the materials and resources that were required for the preparation of this report.

We would take all the responsibility for our possible mistakes in our research report and we will try to resolve them.

Mahesh Bhandari
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TABLE OF CONTENTS

1: INTRODUCTION

1.1 Background	5
1.2 Problem Statement	5
1.3 Objectives of Project	5
1.4 Definition of Technologies Used	6

2: DBMS FOR CV MANAGEMENT SYSTEM

2.1 Entity-Relationship Model	7
2.1.1 Entities and Entity Set	7
2.1.2 Attributes	7
2.1.3 Relationships	7
2.2 E-R Diagram	8
2.3 Attributes Table	8-9
2.4 ER Diagram (Graphical Representation)	10

3: MAPPING AND NORMALIZATION

3.1 ER to Relational Mapping	11-15
------------------------------------	-------

4: DATABASE CREATION

4.1 Data Definition Language (DDL)	16-19
4.2 DataBase Tables	20
4.3 Data Insertion	20-23

5: CONCLUSION

6: REFERENCES	25
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Introduction

1.1 Background

In the current professional landscape, managing a large volume of resumes (CVs) manually is a daunting task for HR departments and recruitment agencies. Traditional methods—such as storing Word documents or PDFs in nested folders—result in significant data silos. Without a structured Database Management System (DBMS), searching for a specific candidate based on a niche skill set or a specific range of experience becomes a time-consuming manual process. This project proposes a **CV Management Database System** to digitize, structure, and streamline the recruitment workflow.

1.2 Problem Statement

- **Data Redundancy:** Storing multiple versions of the same resume leads to wasted storage and confusion.
- **Difficulty in Data Retrieval:** Finding candidates with specific qualifications (e.g., "5+ years in Java") requires opening every file manually.
- **Inefficient utilization of manpower:** Not maintaining a DBMS can lead to inefficient utilization of manpower, as staff may need to spend more time on manual data entry and management tasks, reducing their productivity and preventing them from focusing on other important tasks like interviewing and talent acquisition.

1.3 Objectives of Project

The main objective of this project is to provide proper solutions to the existing problems in candidate data management due to traditional approaches. This project can be used for:

- **To maintain a centralized database** for all candidate profiles and professional histories.
- **To provide proper security of data** by restricting access to sensitive contact information.

- **Efficient data management** through the use of relational tables.
- **To update data on a regular basis**, allowing candidates to add new skills or experiences easily.
- **Easy retrieval of data** using optimized SQL queries for recruitment filtering.

1.4 Definition of Technologies Used

The project uses the following technologies:

- **DBMS: MySQL** SQL stands for Structured Query Language. SQL is the most common standardized language used to access databases. MySQL, the most popular Open Source SQL database management system, is developed, distributed, and supported by Oracle Corporation. The MySQL Database Server is very fast, reliable, scalable, and easy to use.
- **XAMPP**: XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends, consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages. The goal of XAMPP is to build an easy-to-install distribution for developers to get into the world of Apache and MySQL.
- **PhpMyAdmin**: PhpMyAdmin is a free and open-source administration tool for MySQL and MariaDB. As a portable web application written primarily in PHP, it has become one of the most popular MySQL administration tools, especially for web hosting services and local development environments.

2: DBMS for CV Management System

We need to describe the relationship among the entities before preparation of the database. For it, we need to create an entity relationship model which makes it easier to understand the relationship among the entities.

2.1 Entity-Relationship Model

The E-R model is one of the most popular conceptual models which is used for designing a database. It views the real world as a set of basic objects, their characteristics, and associations among these objects. The entities, attributes, and relationships are the basic constructs of an E-R model.

2.1.1 Entities and Entity Set

An entity is an object or a concept that is defined by the enterprise as having an independent existence. In this project, an entity could be a User, a Skill, or an Experience record. An entity usually has an attribute whose value is distinct for each individual entity in the collection. An entity set is a set of entities of the same type that share the same properties, or attributes.

2.1.2 Attributes

Attributes are the information that explains the properties of an entity. Attributes are the descriptive properties possessed by all members of an entity set. Each entity may have its own value for each attribute (e.g., a User entity has a "First Name" and "Email"). Attributes are represented by ovals with the attribute name.

2.1.3 Relationships

Association between two or more entities is called a relationship. In the CV database, a "User" is associated with "Education." So the entity takes part in the relationship, and it is represented by a diamond shape.

2.2 E-R Diagram

Once the entity types, relationship types, and their corresponding attributes have been identified, the next step is to graphically represent these components using an E-R diagram. An E-R diagram is a specialized graphical tool that demonstrates the interrelationships among various entities of a database. It is used to represent the overall logical structure of the database.

Requirements

In the ER model, the first step is to identify the entities of the organization that we are interested in.

Entities of CV Management System

An entity is a thing or object in the real world that is distinguishable from other objects. In the case of the CV Management System, five core entities are identified. They are:

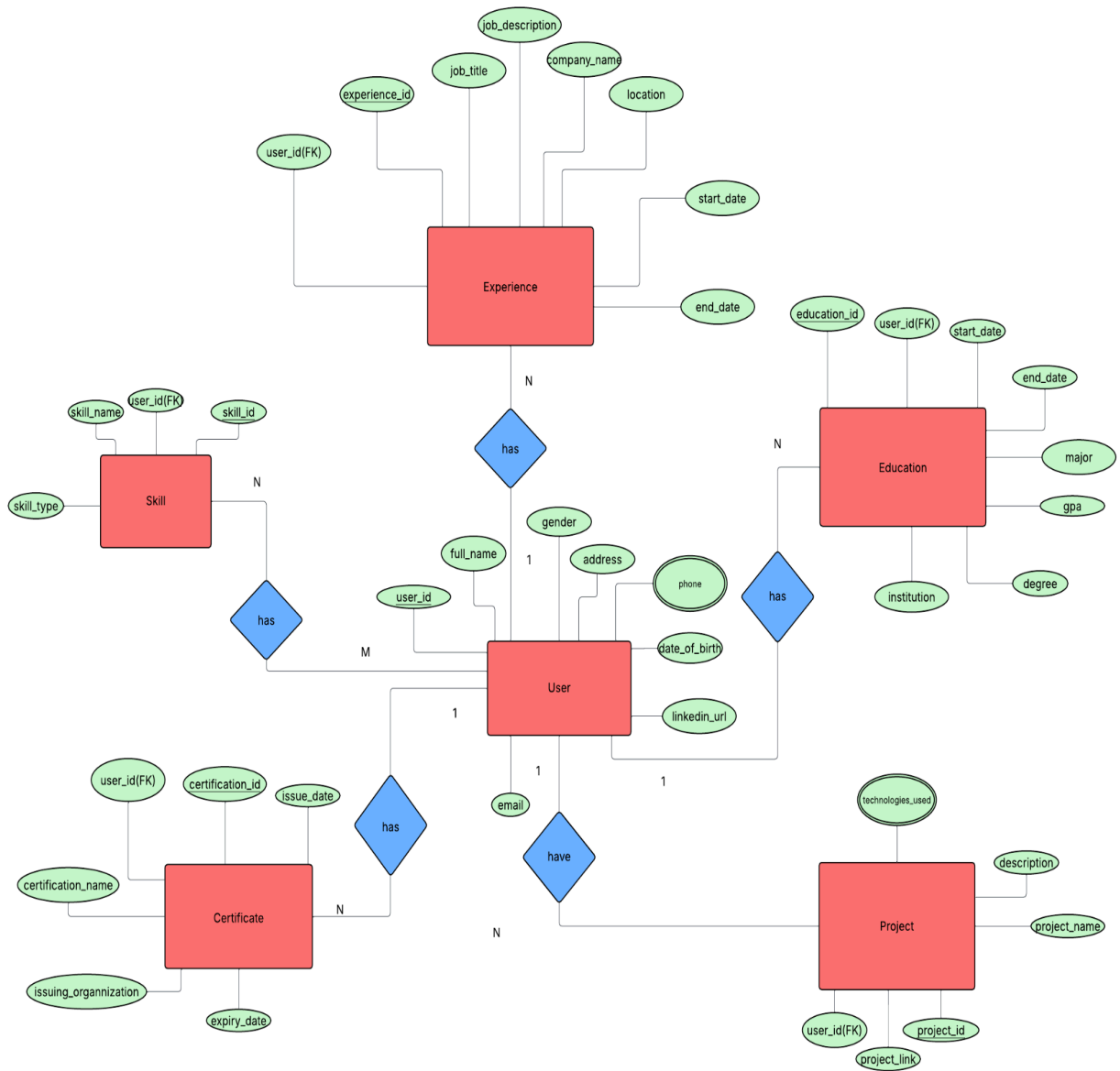
1. **User** (The job seeker/profile owner)
2. **Education** (Academic background)
3. **Experience** (Professional work history)
4. **Skill** (Technical or soft skills)
5. **Project** (Portfolios or specific works)
6. **Certification** (Certificates of different works)

2.3 Attributes Table

An entity is represented by a set of attributes. The attribute(s) which can uniquely identify the Entity is called the Primary Key. The main Attributes of all the above Entities are listed in the table below. The Primary Keys are underlined.

Entities	Attributes
User	<u>user_id</u> , full_name, email, phone, address, linkedin_url,gender,date_of_birth
Education	<u>education_id</u> ,institution,degree, major,gpa,start_date,end_date
Experience	<u>experience_id</u> , company_name, job_title, start_date, end_date,job_description,location
Skill	<u>skill_id</u> , skill_name, skill_type
Project	<u>project_id</u> , project_name, description, project_link,technologies_used
Certificate	<u>certification_id</u> ,certification_name,issue_date,expiry_date, issuing_organization

2.4 ER Diagram

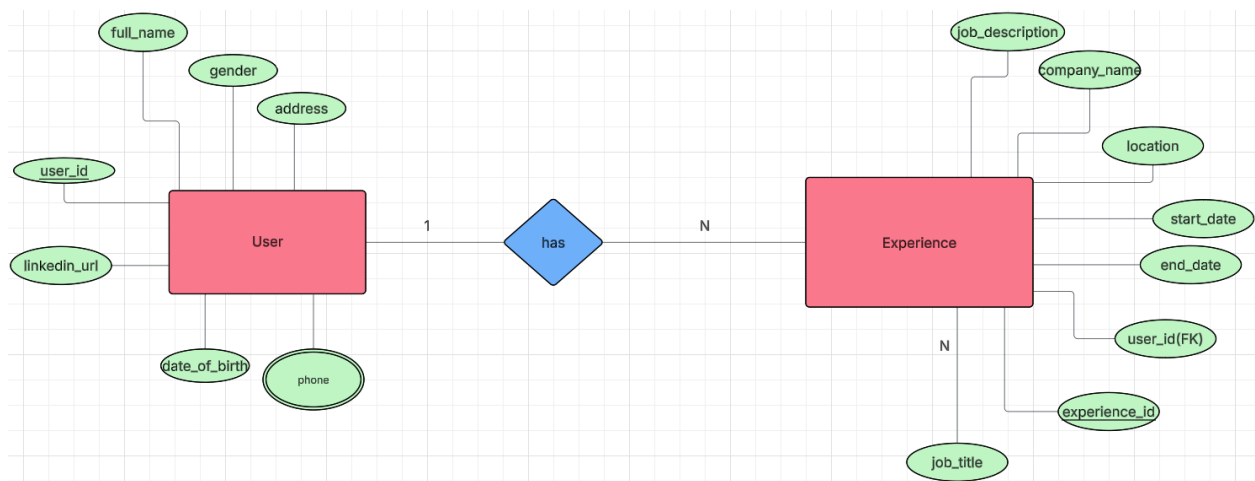


3 : Mapping ER Model To Relational Model and Normalization

We ensured that during the mapping of the ER diagram to the relational model, the database was normalized up to Third Normal Form (3NF) to eliminate redundancy and maintain data integrity.

Step 1: User and Experience

There is a 'One to Many' Relationship between User and Experience. So, we use **user_id** as a **Foreign Key** in the experience table.



User Table:-

<u>user_id</u>	gender	full_name	address	linkedin_url	date_of_birth
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User-phone table:-

Since a user can have multiple phone numbers, a separate User_phone table is created to handle the multivalued attribute phone. It contains user_id as a foreign key referencing the User table.

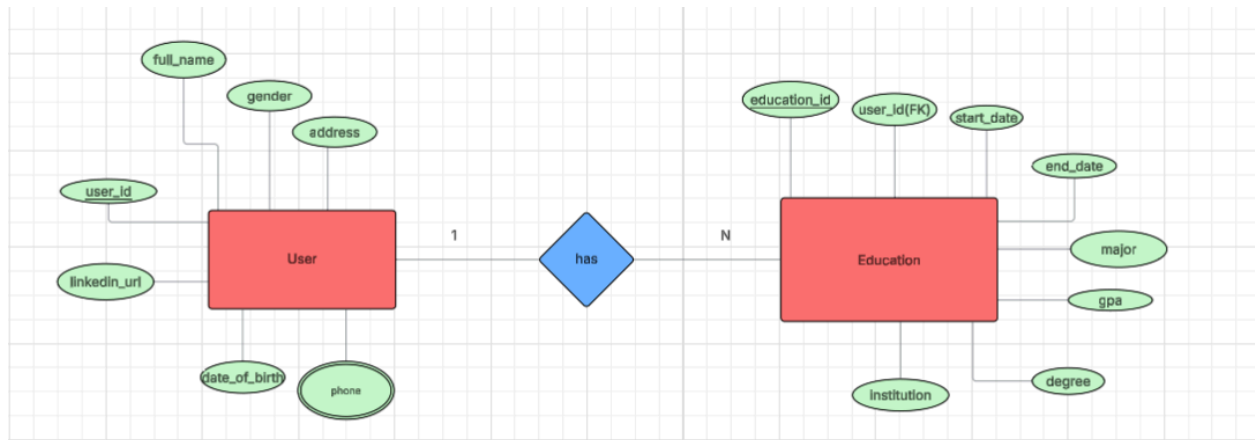
<u>user_id</u> (FK)	<u>phone</u>
---------------------	--------------

Experience table:-

<u>experience_id</u>	job_title	job_description	company_name	location	start_date	end_date	user_id (FK)
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Step 2: Education:-

There is a 'One to Many' Relationship between User and Education. So, we use **user_id** as a **Foreign Key** in the education table.

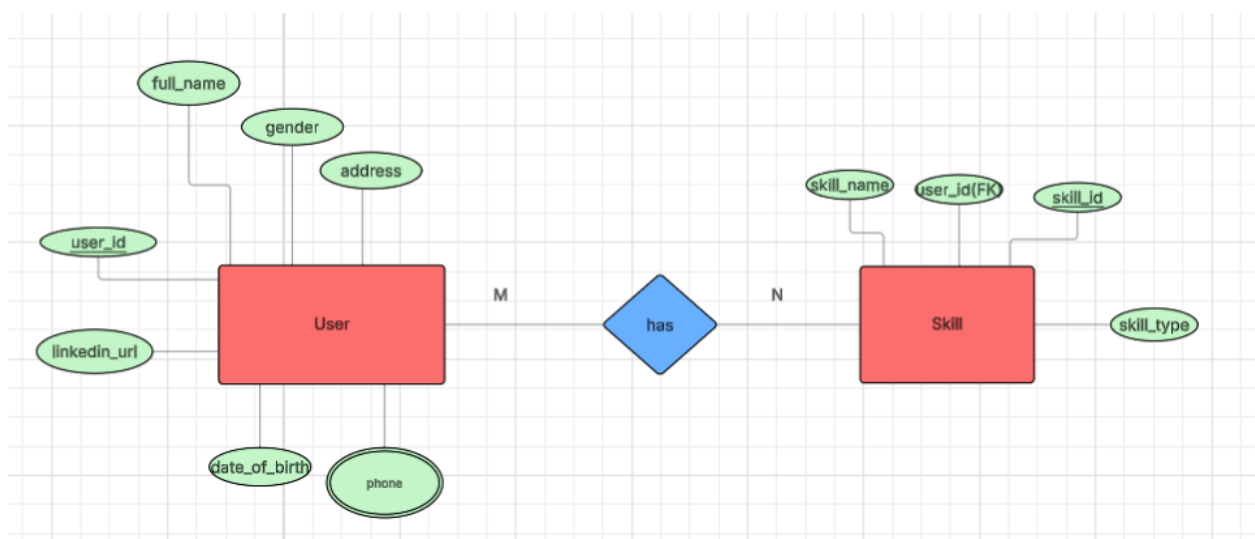


Education Table:-

<u>education_id</u>	major	gpa	degree	institution	start_date	end_date	user_id(FK)
---------------------	-------	-----	--------	-------------	------------	----------	-------------

Step 3: Skill:-

There is a 'Many to Many' Relationship between the user and skill. So, we will create a new table for **has** Relationship. So, we will create user_skill table that can store the data for many to many relationship.



Skill Table:-

<u>skill_id</u>	skill_name	skill_type
-----------------	------------	------------

In this table we create a skill table where we store the skills along with its types.

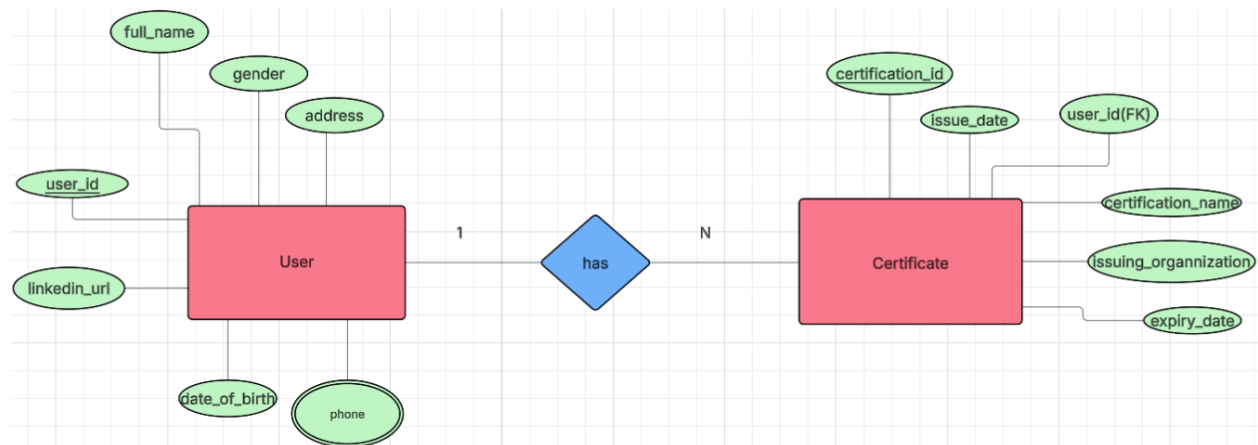
User_skill Table:-

<u>user_id</u> (FK)	<u>skill_id</u> (FK)	proficiency_level
---------------------	----------------------	-------------------

Here, both user_id and skill_id both acts as a candidate key and tells the information about which user has which skill and so on along with their proficiency_level. It also ensures that normalization is maintained up to 3NF.

Step 4: Certificate:-

There is a 'One to Many' Relationship between User and Certificate. So, we use **user_id** as a **Foreign Key** in the education table.

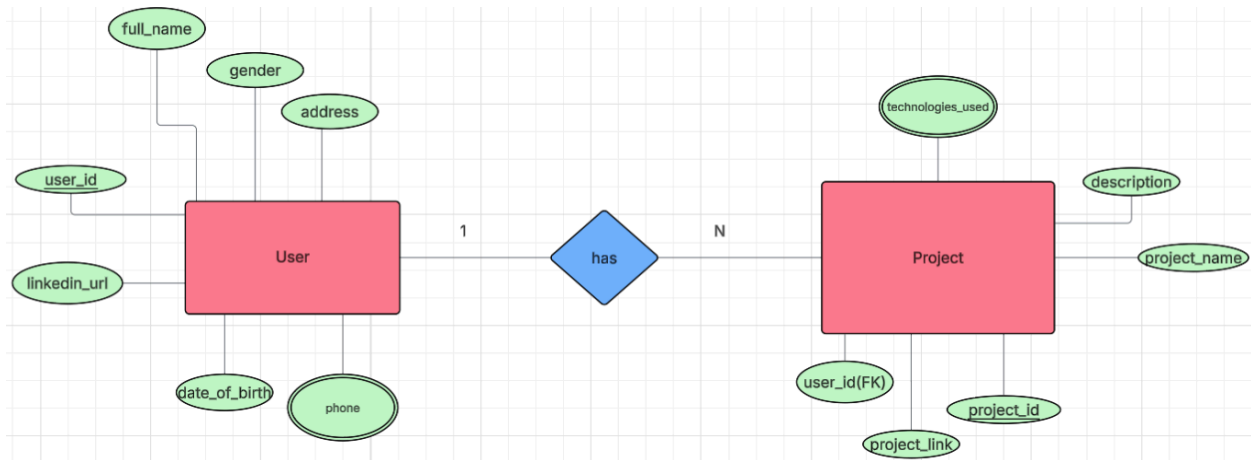


Certificate Table:-

<u>certification_id</u>	certification_name	issuing_organization	issue_date	expiry_date	user_id(FK)
-------------------------	--------------------	----------------------	------------	-------------	-------------

Step 5: Project:-

There is a 'One to Many' Relationship between User and Project. So, we use **user_id** as a **Foreign Key** in the experience table.



Project Table:-

<u>project_id</u>	project_name	project_link	description	user_id(FK)
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Project_technology Table:-

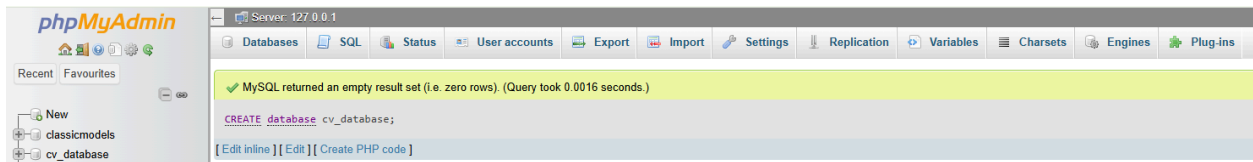
Since a project can have multiple technologies used so separate project_technology table is created to handle the multivalued attribute. It contains project_id as a foreign key referencing the project table. It is also done to maintain normalization up to 3NF.

<u>project_id</u> (FK)	<u>technology_name</u>
------------------------	------------------------

The ER diagram of the CV database was successfully transformed into a relational model by converting each entity into a table and representing relationships using primary and foreign keys. Multivalued attributes were handled through separate relations, and the one-to-many relationships were maintained using foreign key constraints. During this mapping process, normalization was carefully applied up to Third Normal Form (3NF) to eliminate redundancy, prevent update anomalies, and ensure data consistency. As a result, the final relational schema is well-structured, efficient, and maintains strong referential integrity.

4 : Database Creation

At first, database was created using SQL command `CREATE DATABASE CV_Management_System` which is shown below:



4.1 Table Creation

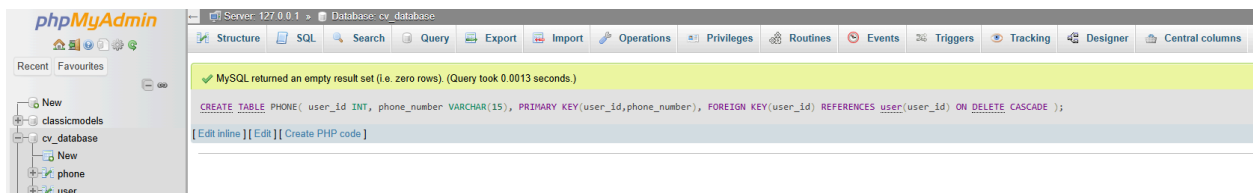
After creating the database, the second step is to create a table using SQL command **CREATE TABLE**. While creating table in almost all tables we have write one common command i.e ON DELETE CASCADE which means if user is removed from database then its corresponding data in another table will also removed. The table creation commands used in the report are shown below:

4.1.1 User Table:-



#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 user_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 full_name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 date_of_birth	date			No	None			Change Drop More
☐	4 email	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	5 address	varchar(200)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 gender	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
☐	7 linkedin_url	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More

4.1.2 Phone Table:-



#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 user_id	int(11)			No	None			Change Drop More
☐	2 phone_number	varchar(15)	utf8mb4_general_ci		No	None			Change Drop More

4.1.3 Project Table:-

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 project_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	int(11)			No	None			Change Drop More
☐	3 project_name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 description	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 project_link	varchar(1000)	utf8mb4_general_ci		Yes	NULL			Change Drop More

4.1.4 Project Technology Table:-

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 project_id	int(11)			No	None			Change Drop More
☐	2 technology_name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

4.1.5 Skill Table:-

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 skill_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 skill_name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 skill_type	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More

4.1.6 User_Skill Table:-

```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0011 seconds)

CREATE TABLE user_skill( user_id INT, skill_id INT, proficiency_level VARCHAR(50), PRIMARY KEY(user_id,skill_id), FOREIGN KEY(user_id) REFERENCES user(user_id) ON DELETE CASCADE, FOREIGN KEY(skill_id) REFERENCES skill(skill_id) ON DELETE CASCADE );

[ Edit inline ] [ Edit ] [ Create PHP code ]

```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 user_id	int(11)			No	None			Change Drop More
☐	2 skill_id	int(11)			No	None			Change Drop More
☐	3 proficiency_level	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More

4.1.7 Education Table:-

```

Server: 127.0.0.1 • Database: cv_database • Table: user_skill

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0013 seconds)

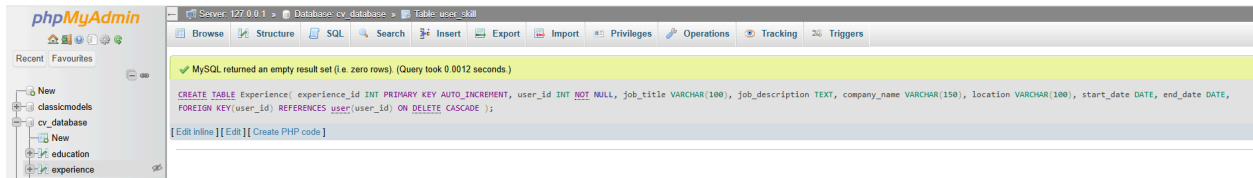
CREATE TABLE Education( education_id INT PRIMARY KEY AUTO_INCREMENT, user_id INT NOT NULL, degree VARCHAR(100), major VARCHAR(100), institution VARCHAR(150), start_date YEAR, end_date YEAR, gpa DECIMAL(3,2), FOREIGN KEY (user_id) REFERENCES user(user_id) ON DELETE CASCADE );

[ Edit inline ] [ Edit ] [ Create PHP code ]

```

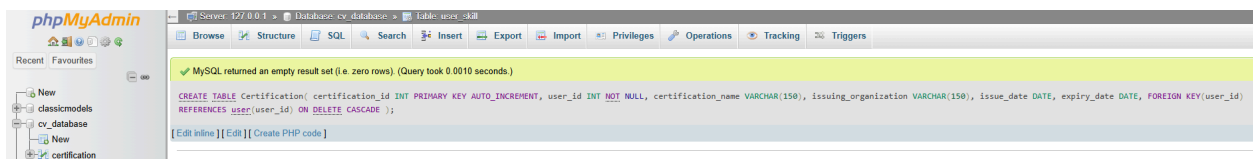
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 education_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	int(11)			No	None			Change Drop More
☐	3 degree	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 major	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 institution	varchar(150)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 start_date	year(4)			Yes	NULL			Change Drop More
☐	7 end_date	year(4)			Yes	NULL			Change Drop More
☐	8 gpa	decimal(3,2)			Yes	NULL			Change Drop More

4.1.8 Experience Table:-



#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 experience_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	int(11)			No	None			Change Drop More
☐	3 job_title	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 job_description	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 company_name	varchar(150)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 location	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	7 start_date	date			Yes	NULL			Change Drop More
☐	8 end_date	date			Yes	NULL			Change Drop More

4.1.9 Certification Table:-



#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 certification_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	int(11)			No	None			Change Drop More
☐	3 certification_name	varchar(150)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 issuing_organization	varchar(150)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 issue_date	date			Yes	NULL			Change Drop More
☐	6 expiry_date	date			Yes	NULL			Change Drop More

4.2 DataBase Tables:-

Table	Action	Rows	Type	Collation	Size	Overhead
☐ certification	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ education	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ experience	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ phone	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ project	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ project_technology	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ skill	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ user	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ user_skill	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
9 tables	Sum	0	InnoDB	utf8mb4_general_ci	256.0 KiB	0 B

4.3 Data Insertion:-

4.3.1 Inserting in User Table:-

```
5 rows inserted.
Inserted row id: 5 (Query took 0.0136 seconds.)

INSERT INTO user (full_name, date_of_birth, email, address, gender, linkedin_url) VALUES ('Darpan Giri', '2003-05-14', 'darpangiri5@gmail.com', 'Kathmandu', 'Male', 'https://linkedin.com/in/darpangiri'), ('Aarav Sharma', '2002-08-21', 'aaravsharma2@gmail.com', 'Pokhara', 'Male', 'https://linkedin.com/in/aaravsharma'), ('Sita Karki', '2004-01-10', 'sitakarki0@gmail.com', 'Lalitpur', 'Female', 'https://linkedin.com/in/sitakarki'), ('Rohan Thapa', '2001-11-30', 'rohanthapa5@gmail.com', 'Biratnagar', 'Male', 'https://linkedin.com/in/rohanthapa'), ('Priya Rai', '2003-03-25', 'priyairai@gmail.com', 'Dharan', 'Female', 'https://linkedin.com/in/priyairai');
```

	user_id	full_name	date_of_birth	email	address	gender	linkedin_url
☐ Edit Copy Delete	1	Darpan Giri	2003-05-14	darpangiri5@gmail.com	Kathmandu	Male	https://linkedin.com/in/darpangiri
☐ Edit Copy Delete	2	Aarav Sharma	2002-08-21	aaravsharma2@gmail.com	Pokhara	Male	https://linkedin.com/in/aaravsharma
☐ Edit Copy Delete	3	Sita Karki	2004-01-10	sitakarki0@gmail.com	Lalitpur	Female	https://linkedin.com/in/sitakarki
☐ Edit Copy Delete	4	Rohan Thapa	2001-11-30	rohanthapa5@gmail.com	Biratnagar	Male	https://linkedin.com/in/rohanthapa
☐ Edit Copy Delete	5	Priya Rai	2003-03-25	priyairai@gmail.com	Dharan	Female	https://linkedin.com/in/priyairai

4.3.2 Inserting in Phone Table:-

```
5 rows inserted. (Query took 0.0035 seconds.)

INSERT INTO phone (user_id, phone_number) VALUES (1, '9811111111'), (1, '9822222222'), (2, '9833333333'), (3, '9844444444'), (4, '9855555555');
```

		user_id	phone_number
☐	Edit Copy Delete	1	9811111111
☐	Edit Copy Delete	1	9822222222
☐	Edit Copy Delete	2	9833333333
☐	Edit Copy Delete	3	9844444444
☐	Edit Copy Delete	4	9855555555

4.3.3 Inserting in Project Table:-

✓ 5 rows inserted.
Inserted row id: 15 (Query took 0.0053 seconds)

```
INSERT INTO project (user_id, project_name, description, project_link) VALUES (1, 'Plant classification model', 'Deep learning model to identify disease in plant.', 'https://github.com/darpangiri/plantclassify'), (2, 'E-commerce Website', 'Full-stack e-commerce website.', 'https://github.com/aaravsharma/ecommerce-site'), (3, 'Student Management System', 'Web-based system to manage student records.', 'https://github.com/sitakarki/ecommerce-site'), (4, 'Portfolio Website', 'Personal portfolio website.', 'https://rohanthapa.dev'), (5, 'Sales Prediction', 'Machine learning model to predict sales.', 'https://github.com/priyara/sales-prediction');
```

[Edit inline] [Edit] [Create PHP code]

		project_id	user_id	project_name	description	project_link
☐	Edit Copy Delete	11	1	Plant classification model	Deep learning model to identify disease in plant.	https://github.com/darpangiri/plantclassify
☐	Edit Copy Delete	12	2	E-commerce Website	Full-stack e-commerce website.	https://github.com/aaravsharma/ecommerce-site
☐	Edit Copy Delete	13	3	Student Management System	Web-based system to manage student records.	https://github.com/sitakarki/ecommerce-site
☐	Edit Copy Delete	14	4	Portfolio Website	Personal portfolio website.	https://rohanthapa.dev
☐	Edit Copy Delete	15	5	Sales Prediction	Machine learning model to predict sales.	https://github.com/priyara/sales-prediction

4.3.4 Inserting in Project_technology Table:-

✓ 7 rows inserted (Query took 0.0065 seconds)

```
INSERT INTO project_technology (project_id, technology_name) VALUES (11, 'Tensorflow'), (11, 'Scikit learn'), (12, 'React'), (12, 'Express'), (13, 'Javascript'), (14, 'Nextjs'), (15, 'Scikit learn');
```

[Edit inline] [Edit] [Create PHP code]

		project_id	technology_name
☐	Edit Copy Delete	11	Scikit learn
☐	Edit Copy Delete	11	Tensorflow
☐	Edit Copy Delete	12	Express
☐	Edit Copy Delete	12	React
☐	Edit Copy Delete	13	Javascript
☐	Edit Copy Delete	14	Nextjs
☐	Edit Copy Delete	15	Scikit learn

4.3.5 Inserting in Skill Table:-

```
✓ 10 rows inserted.
Inserted row id: 10 (Query took 0.0036 seconds)

INSERT INTO skill (skill_name, skill_type) VALUES ('C++', 'Programming Language'), ('Python', 'Programming Language'), ('JavaScript', 'Programming Language'), ('MySQL', 'Database'), ('PostgreSQL', 'Database'), ('HTML', 'Frontend'), ('CSS', 'Frontend'), ('React', 'Frontend Framework'), ('Node.js', 'Backend Framework'), ('TensorFlow', 'Machine Learning');
```

		skill_id	skill_name	skill_type
☐	Edit Copy Delete	1	C++	Programming Language
☐	Edit Copy Delete	2	Python	Programming Language
☐	Edit Copy Delete	3	JavaScript	Programming Language
☐	Edit Copy Delete	4	MySQL	Database
☐	Edit Copy Delete	5	PostgreSQL	Database
☐	Edit Copy Delete	6	HTML	Frontend
☐	Edit Copy Delete	7	CSS	Frontend
☐	Edit Copy Delete	8	React	Frontend Framework
☐	Edit Copy Delete	9	Node.js	Backend Framework
☐	Edit Copy Delete	10	TensorFlow	Machine Learning

4.3.6 Inserting in User_skill Table:-

```
✓ 9 rows inserted. (Query took 0.0074 seconds)

INSERT INTO user_skill (user_id, skill_id, proficiency_level) VALUES (1, 1, 'Advanced'), (1, 2, 'Intermediate'), (1, 4, 'Advanced'), (2, 3, 'Advanced'), (2, 8, 'Intermediate'), (3, 6, 'Advanced'), (3, 7, 'Advanced'), (4, 9, 'Intermediate'), (5, 10, 'Beginner');
```

		user_id	skill_id	proficiency_level
☐	Edit Copy Delete	1	1	Advanced
☐	Edit Copy Delete	1	2	Intermediate
☐	Edit Copy Delete	1	4	Advanced
☐	Edit Copy Delete	2	3	Advanced
☐	Edit Copy Delete	2	8	Intermediate
☐	Edit Copy Delete	3	6	Advanced
☐	Edit Copy Delete	3	7	Advanced
☐	Edit Copy Delete	4	9	Intermediate
☐	Edit Copy Delete	5	10	Beginner

4.3.7 Inserting in Experience Table:-

```
5 rows inserted
Inserted row id: 5 (Query took 0.0053 seconds.)

INSERT INTO experience (user_id, job_title, job_description, company_name, location, start_date, end_date) VALUES (1, 'Backend Developer Intern', 'REST API development.', 'Tech Solutions Pvt Ltd', 'Kathmandu', '2024-01-01', '2024-06-30'), (2, 'Frontend Developer', 'Developed responsive UI using React .', 'Web Innovators', 'Pokhara', '2023-03-15', '2024-02-15'), (3, 'Software Engineer', 'Built student management systems and handled database integration.', 'EduTech Nepal', 'Lalitpur', '2022-07-01', '2023-12-31'), (4, 'Full Stack Developer', 'Worked on MERN stack applications.', 'NextGen IT', 'Biratnagar, Nepal', '2023-01-10', '2023-06-06'), (5, 'Machine Learning Intern', 'Developed predictive models for sales forecasting.', 'Data Analytics Hub', 'Chitwan, Nepal', '2024-05-01', '2024-10-01');
```

	experience_id	user_id	job_title	job_description	company_name	location	start_date	end_date
☐ Edit ☐ Copy ☐ Delete	1	1	Backend Developer Intern	REST API development.	Tech Solutions Pvt Ltd	Kathmandu	2024-01-01	2024-06-30
☐ Edit ☐ Copy ☐ Delete	2	2	Frontend Developer	Developed responsive UI using React .	Web Innovators	Pokhara	2023-03-15	2024-02-15
☐ Edit ☐ Copy ☐ Delete	3	3	Software Engineer	Built websites and handled database integration.	EduTech Nepal	Lalitpur	2022-07-01	2023-12-31
☐ Edit ☐ Copy ☐ Delete	4	4	Full Stack Developer	Worked on MERN stack applications.	NextGen IT	Biratnagar, Nepal	2023-01-10	2023-06-06
☐ Edit ☐ Copy ☐ Delete	5	5	Machine Learning Intern	Developed predictive models for sales forecasting.	Data Analytics Hub	Chitwan, Nepal	2024-05-01	2024-10-01

4.3.8 Inserting in Certification Table:-

```
5 rows inserted
Inserted row id: 5 (Query took 0.0047 seconds.)

INSERT INTO certification (user_id, certification_name, issuing_organization, issue_date, expiry_date) VALUES (1, 'MySQL Database Certification', 'Oracle', '2024-06-15', '2027-06-15'), (2, 'Frontend Development with React', 'Coursera', '2023-09-10', NULL), (3, 'Python for Data Science', 'Udemy', '2024-01-20', NULL), (4, 'Full Stack Web Development', 'edX', '2023-05-05', '2026-05-05'), (5, 'Machine Learning Specialization', 'Coursera', '2024-08-01', NULL);
```

	certification_id	user_id	certification_name	issuing_organization	issue_date	expiry_date
☐ Edit ☐ Copy ☐ Delete	1	1	MySQL Database Certification	Oracle	2024-06-15	2027-06-15
☐ Edit ☐ Copy ☐ Delete	2	2	Frontend Development with React	Coursera	2023-09-10	NULL
☐ Edit ☐ Copy ☐ Delete	3	3	Python for Data Science	Udemy	2024-01-20	NULL
☐ Edit ☐ Copy ☐ Delete	4	4	Full Stack Web Development	edX	2023-05-05	2026-05-05
☐ Edit ☐ Copy ☐ Delete	5	5	Machine Learning Specialization	Coursera	2024-08-01	NULL

4.3.9 Inserting in Education Table:-

```
5 rows inserted
Inserted row id: 5 (Query took 0.0056 seconds.)

INSERT INTO education (user_id, degree, major, institution, start_date, end_date, gpa) VALUES (1, 'Bachelor of Engineering', 'Computer Engineering', 'Tribhuvan University', 2022, 2026, 3.75), (2, 'Bachelor of Science', 'Information Technology', 'Pokhara University', 2021, 2025, 3.60), (3, 'Bachelor of Computer Applications', 'Software Development', 'Kathmandu University', 2020, 2024, 3.80), (4, 'Bachelor of Engineering', 'Electronics and Communication', 'Purbanchal University', 2019, 2023, 3.45), (5, 'Bachelor of Science', 'Data Science', 'Mid-Western University', 2022, 2026, 3.90);
```

	education_id	user_id	degree	major	institution	start_date	end_date	gpa
☐ Edit ☐ Copy ☐ Delete	1	1	Bachelor of Engineering	Computer Engineering	Tribhuvan University	2022	2026	3.75
☐ Edit ☐ Copy ☐ Delete	2	2	Bachelor of Science	Information Technology	Pokhara University	2021	2025	3.60
☐ Edit ☐ Copy ☐ Delete	3	3	Bachelor of Computer Applications	Software Development	Kathmandu University	2020	2024	3.80
☐ Edit ☐ Copy ☐ Delete	4	4	Bachelor of Engineering	Electronics and Communication	Purbanchal University	2019	2023	3.45
☐ Edit ☐ Copy ☐ Delete	5	5	Bachelor of Science	Data Science	Mid-Western University	2022	2026	3.90

5: Conclusion

In conclusion, a database management system (DBMS) is a crucial tool for efficiently managing professional data within the CV Management System. It provides HR professionals and recruitment agencies with a structured way to store, organize, retrieve, and manipulate candidate profiles easily and efficiently. With a DBMS, users can reduce data redundancy—such as duplicate resume entries—and improve data accuracy, consistency, and integrity across multiple candidate records. It also enables multiple recruiters to access and filter the candidate data simultaneously, ensuring data security and confidentiality of sensitive contact information.

However, DBMS implementation for a CV system requires careful planning and design, particularly in the normalization of skills and work history, to ensure that it meets the requirements of the organization and its hiring goals. The choice of MySQL as the DBMS was based on factors such as scalability for large candidate pools, performance in high-speed querying, availability, and cost-effectiveness. Regular maintenance and optimization of the database tables and indexes are also essential to ensure its optimal performance and prevent data loss or corruption of vital professional records.

In summary, a DBMS is a powerful tool that can significantly improve an organization's talent acquisition capabilities and enhance their decision-making processes when selecting the right candidate. It is an essential component of a modern HR information system, and its efficient management can lead to a competitive advantage in the marketplace by identifying the best talent faster and more accurately than traditional manual methods.

6: References

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